

SOFTWARE PROJECT MANAGEMENT

MCA 5063

PE--III

Introduction



- 1. How many of you have worked on a s/w project that was completed on time within the budget without changing any requirements?**
- 2. Why do they get delayed?**
- 3. Scenario – E-learning Management System**
Reqd: Student login, course regn and assignment submission
After 4 months added: attendance, video lecture capture, online exams
2 developers left, budget and time remain same in this scenario as a project manager what will you do?

Options

Why projects fail?

Technical problems:

Management Problems:



Course Objectives

- 1. Introduce the fundamental concepts of software project management and development processes.**
- 2. Provide knowledge of project planning techniques and requirement change management.**
- 3. Introduce quality assurance, risk management, configuration management, and measurement practices.**
- 4. Provide exposure to project management plans, organizational process assets, project monitoring and control**
- 5. Study frameworks for process improvement, project maturity, and organizational knowledge management.**

Course outcomes

- 1. Analyze software project management principles, development processes, and stakeholder requirements for effective project planning.**
- 2. Apply estimation, scheduling, and resource planning techniques to effectively design and manage project activities within cost and time constraints.**
- 3. Develop and evaluate risk, quality, and configuration management strategies to ensure process integrity and control in software projects.**
- 4. Implement monitoring control and corrective action mechanisms to address schedule, cost variances while maintaining project objectives.**
- 5. Use frameworks for process improvement, project maturity, and organizational knowledge management throughout the project lifecycle.**

Reference Books

- 1. Pankaj Jalote, Software Project Management in Practice, 1st Edition, Addison, Wesley, 2014.**
- 2. Bob Hughes, Mike Cotterell, Rajib Mall, Software Project Management, 6th Edition McGraw Hill Education, 2017.**
- 3. Roger S. Pressman, Software Engineering: A Practitioner's Approach, 8th Edition, McGraw Hill Publication, 2014.**
- 4. Normal E Fenton, Shari Lawrence, Software Metrics, 5th Edition, Pfleeger Thompson, 2010.**

Assessment

Internal Assessment :

2 Surprise Assignment → 5M each (2X5 → 10M)

1 Project Management Demo on case study → 10M

Mid-Term Exam → 30M

End Semester Exam → 50 M



Topics

- 1. Processes and Project Management**
- 2. Change Management**
- 3. Effort Estimation and Scheduling**
- 4. Quality Planning**
- 5. Risk Management**
- 6. Configuration Management, Measurement and Tracking Planning**
- 7. Project Management Plan**
- 8. Project Monitoring and Control**
- 9. Project Closure**

What is a Software Process?

- **A software process is a structured set of activities, methods, roles, and deliverables used to develop and maintain software.**
- **Typical activities include:**
 - **Requirement analysis**
 - **System design**
 - **Coding**
 - **Testing**
 - **Deployment**
 - **Maintenance**
- **A software process establishes consistency across projects and helps ensure predictable quality and delivery.**



Example :

University E-Learning Management System

Requirements:

Student registration

S/W development Model

SDLC or Any other?

Project Manager preparation

Project initiation, timeline

Plan

Risk register

Communication and Quality plan



What is a Project?

- A project is defined as an **endeavor** undertaken to create a **unique product, service, or result**.

Features:

- **Definitive start and end**
- **Specific objectives**
- **Limited resources**
- **Differentiates a project from routine operations.**

Project	Product	Service	Result
Hospital Management System	✓		
Mobile Banking App	✓		
Food Delivery App	✓		
Online Payment Gateway Service		✓	
Cloud Data Backup Service		✓	
Cybersecurity Monitoring Service		✓	
Software Process Audit			✓

Company	Example	Type
Microsoft	Microsoft Teams application	Product
Google	Gmail cloud email platform	Service
Amazon Web Services (AWS)	Cloud hosting services	Service
Infosys	Software process maturity assessment for a client	Result
TCS	IT consulting and process optimization report	Result
Oracle	Database Management System	Product

Scenario	Expected Answer (Write product/service/result)
Develop a mobile app for online shopping.	
Implement a 24×7 cloud-based technical support platform.	
Conduct an audit of an organization's software development practices and submit recommendations.	
Create a college ERP system.	
Provide a managed cybersecurity monitoring solution.	
Prepare a feasibility study for migrating legacy software to the cloud.	

Why is Software Project Management Important?

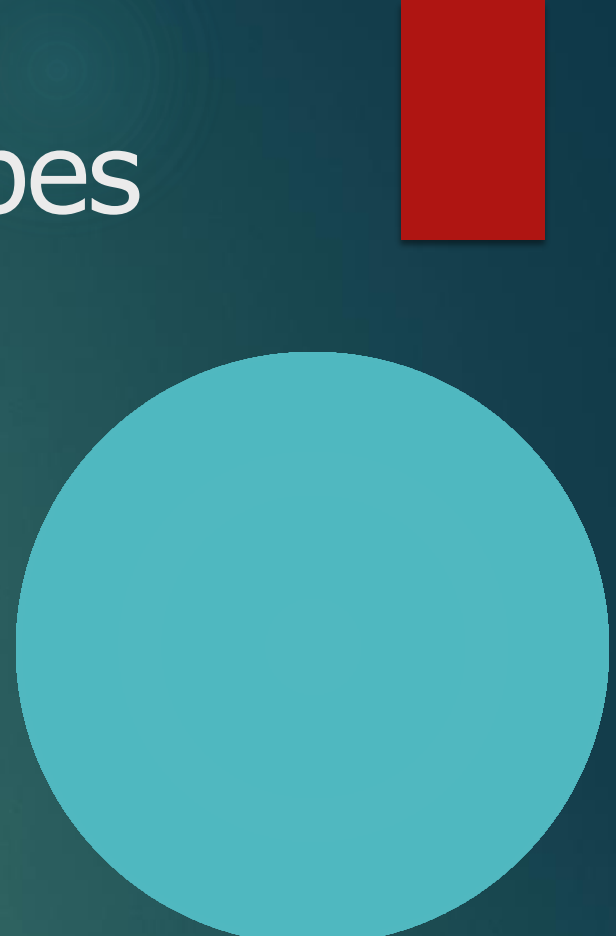
- Software projects are often **complex, innovative, and resource-intensive**.
- Without proper management, projects suffer from **budget overruns, missed deadlines, and quality issues**.
- Emphasis on the increasing reliance of businesses and society on **software systems** in critical domains (finance, healthcare, etc.).
- Examples of software failures (e.g., in air traffic control or e-commerce) highlight the stakes.

Differences Between Software Process and Project Management

	SOFTWARE PROCESS	PROJECT MANAGEMENT
FOCUS		
ACTIVITIES		
DELIVERABLES		
MANAGEMENT		



Software Projects vs. Other Types of Projects



- Software projects are more **abstract and intangible**.
- High **rate of change** in requirements.
- Difficulties in **measurement** and **progress tracking**.
- Team-based intellectual activity, unlike physical construction projects.
- Testing is **extensive and critical**.

Contract Management and Technical Project Management

- Contract Management: Focuses on **terms, conditions, deliverables**, and **legal aspects** of project agreements.
- Technical Project Management: Involves **planning, scheduling, resource allocation, tracking**, and **team coordination**.

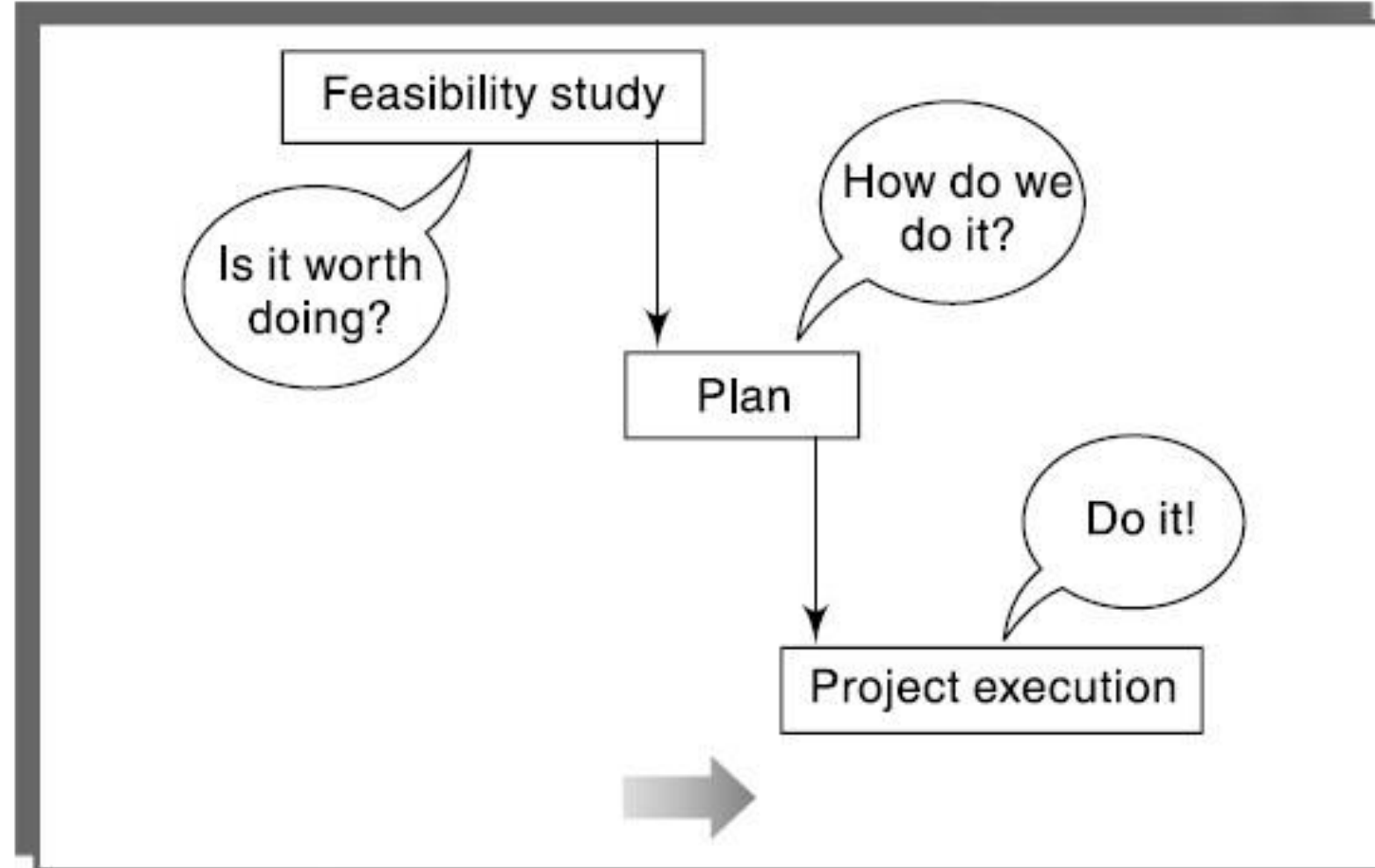
Activities Covered by Software Project Management

SPM includes:

- ▶ **Initiation and planning:** Feasibility, resource allocation, scheduling.
- ▶ **Execution and monitoring:** Tracking progress, quality assurance.
- ▶ **Closure:** Final delivery, documentation, post-project analysis.

Also includes:

- ▶ **Risk management**
- ▶ **Configuration and change management**
- ▶ **Effort estimation**
- ▶ **Quality and resource management**



The feasibility study/plan/execution cycle

Feasibility Type	Analysis Question	Example for E-learning Software
Technical Feasibility	Do we have the required technology and expertise?	The organization evaluates whether it can develop a web-based LMS using technologies such as React for frontend, Java/Spring Boot for backend, and MySQL/PostgreSQL database. Availability of cloud infrastructure (AWS/Azure) is checked for hosting.
Economic Feasibility	Is the project financially beneficial?	Estimated development cost is compared with expected benefits. For example, developing an LMS may cost \$100,000 , but it can reduce classroom training costs and support 10,000 online learners annually.
Operational Feasibility	Will users accept and effectively use the system?	Faculty members and students are surveyed to check whether they are comfortable using online lectures, digital assignments, and online examinations.
Schedule Feasibility	Can the project be completed within the required time?	The university requires the LMS before the next academic semester (6 months). The project manager checks whether analysis, design, development, testing, and deployment can fit within this timeline.

RESOURCE ALLOCATION MATRIX

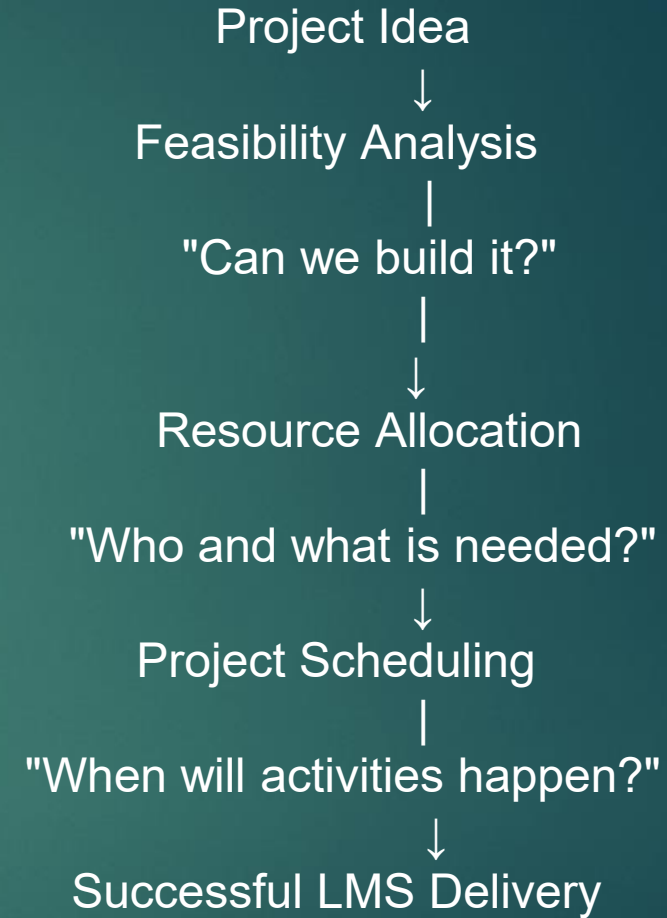
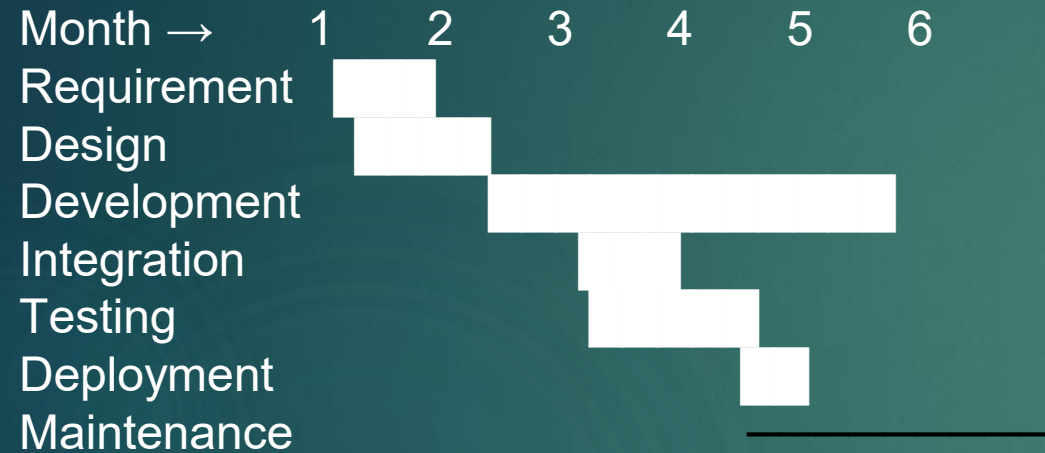
Project Activity	Assigned Resources
Requirement Analysis	Project Manager + Business Analyst + Faculty Representatives
UI Design	UI Designer + Frontend Developer
Database Design	Backend Developer + Database Administrator
Coding	Frontend and Backend Developers
Testing	QA Engineers
Deployment	System Administrator + DevOps Engineer

SCHEDULE FOR E-LEARNING

Phase	Major Activities	Duration	Deliverable
1. Requirement Analysis	Collect requirements from students, faculty, administrators	3 weeks	Software Requirement Specification (SRS)
2. System Design	Database design, architecture design, UI design	4 weeks	Software Design Document
3. Development	Coding of login, course management, video lectures, assessments	12 weeks	Working LMS prototype
4. Integration	Combine modules and integrate payment/video systems	3 weeks	Integrated system
5. Testing	Functional testing, security testing, performance testing	5 weeks	Tested LMS
6. Deployment	Install system on cloud server and train users	2 weeks	Operational LMS
7. Maintenance	Bug fixing and enhancements	Continuous	Updated versions

A project manager may use techniques such as:

- **Work Breakdown Structure (WBS)**
- **Gantt Chart**
- **PERT/CPM Network**
- **Milestone Planning**



Scenario:

"A university wants to develop a mobile-based e-learning application for 20,000 students."

Prepare:

1.Feasibility Analysis

1. Is cloud hosting required?
2. What technology can be used?
3. What are possible risks?

2.Resource Allocation

1. How many developers?
2. What testing team is required?
3. What infrastructure is needed?

3.Scheduling

1. Prepare a 6-month Gantt chart.
2. Identify the critical activities.

ISO/IEC 12207

- **ISO/IEC 12207** is an international standard that defines a **common framework of software life cycle processes**.
- It provides a structured set of processes, activities, and tasks required for the **development, operation, maintenance, and disposal of software systems**.
- It was first published by the **International Organization for Standardization (ISO)** and the **International Electrotechnical Commission (IEC)**.
- The standard does **not prescribe a specific software development methodology** (such as Waterfall, Agile, Spiral, or DevOps). Instead, it defines **what processes should exist** during the software life cycle.

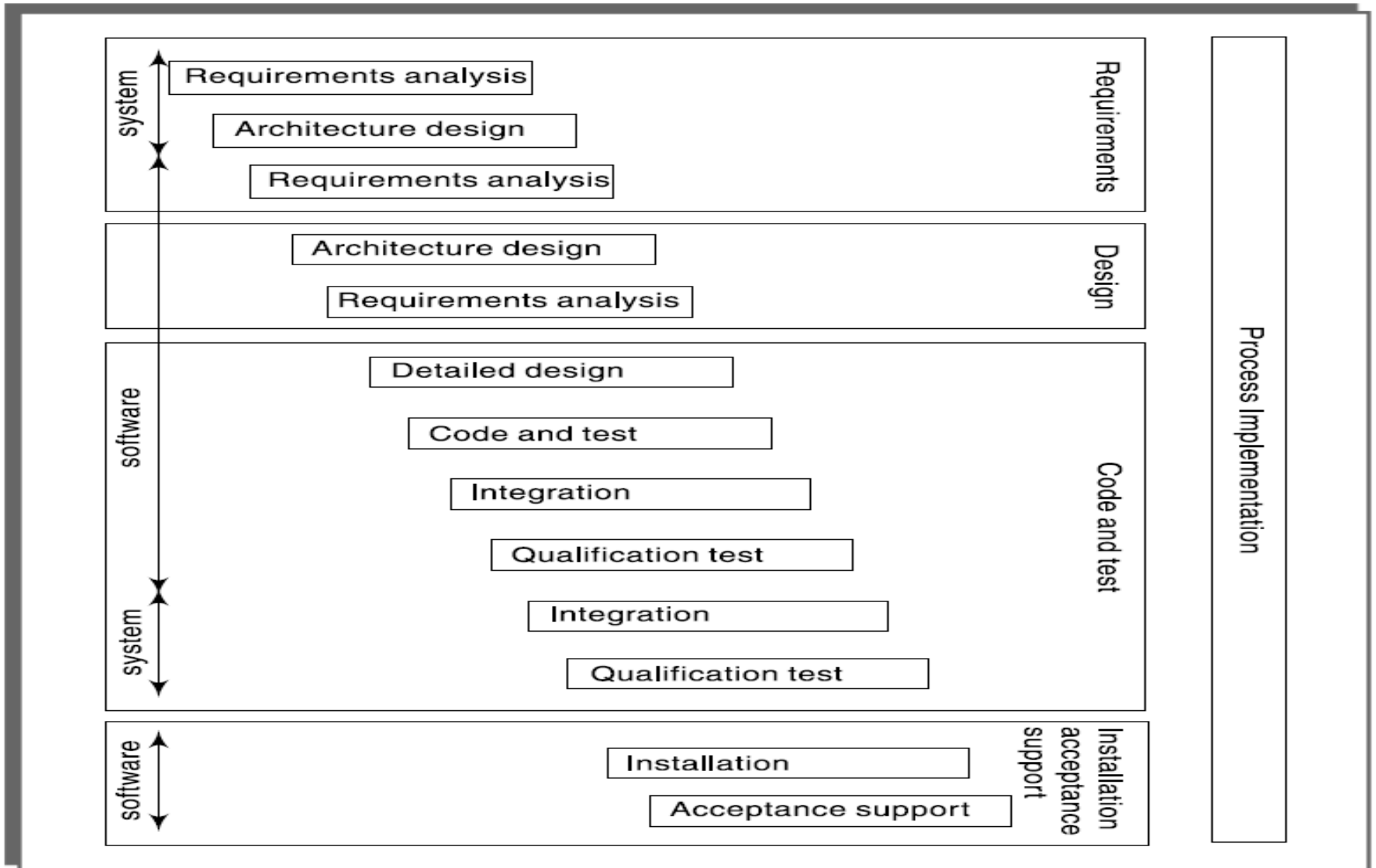


FIGURE 1.3 The ISO 12207 software development life cycle

ISO/IEC 12207 Software Life Cycle Processes

- 1. Primary life cycle processes**
- 2. Supporting life cycle processes**
- 3. Organizational life cycle processes**
- 4. Agreement Processes**



Mapping SDLC and ISO Processes

SDLC Phase	ISO/IEC 12207 Process
Feasibility Study	Acquisition + Project Management
Requirement Analysis	Development Process
System Design	Development Process
Coding	Implementation Process
Testing	Verification + Validation
Deployment	Operation Process
Maintenance	Maintenance Process
Documentation	Supporting Process
Quality Control	Quality Assurance

ISO Process	SCEMS Activity
Acquisition	University selects software vendor
Development	Build mobile app and dashboard
Verification	Perform code and design reviews
Validation	University accepts system
Operation	Security team uses system
Maintenance	Fix bugs and add features
Configuration Management	Manage software versions

	Software Process Management	Software Project Management
Definition		
Key Question Addressed		
Measurement Metrics		
Risk Handling		
Quality Management		
Change Management		
Example		

Some Ways of Categorizing Software Projects

Projects can be categorized based on:

- ▶ **Compulsory vs. Voluntary Users**
- ▶ **Information Systems vs. Embedded Systems**
- ▶ **Outsourced Projects**
- ▶ **Objective-Driven Development**



Compulsory vs. Voluntary Users

- **Compulsory:** Users must use the system (e.g., internal tools like payroll).
- Focus on **functionality, reliability, and compliance.**
- **Example:** Hospital staff using an Electronic Health Records (EHR) system.
- **Voluntary:** Users choose to use the software (e.g., mobile apps).
- UX, performance, and satisfaction are **critical for adoption.**
- **Example:** A meditation app or a personal budgeting tool.

Information Systems vs. Embedded Systems

- **Information Systems:** Handle data, user-facing (e.g., CRM, ERP).
 - Emphasis on **data flow**, **interfaces**, and **usability**.
 - Examples: Inventory management system, Student Information System (SIS), CRM software.
- **Embedded Systems:** Software within hardware (e.g., pacemaker, car ECU).
 - Require **real-time control**, **hardware integration**, and **rigorous testing**.
 - Examples: Software in microwave ovens, pacemakers, automotive control systems.

Outsourced Projects

- ▶ Developed by **external vendors** (offshore/onshore).
- ▶ Involves **contracts, SLAs, and clear deliverables**.
- ▶ Requires strong **communication** and **quality control**.
- ▶ Risk of **misalignment** in goals or expectations.
- ▶ Common in situations where:
 - ▶ **Cost-saving** is important.
 - ▶ **Specialized skills** are needed temporarily.

Objective-Driven Development

- Focuses on achieving **specific business outcomes** (e.g., reduce costs).
- Prioritizes features based on **value delivered**, not just effort.
- Encourages **frequent reviews, metrics, and adaptive planning**.
- Common in **Agile and value-driven frameworks**.

Examples:

- ▶ Redesigning a website to **increase conversions by 20%**.
- ▶ Automating a workflow to **reduce processing time by 40%**.

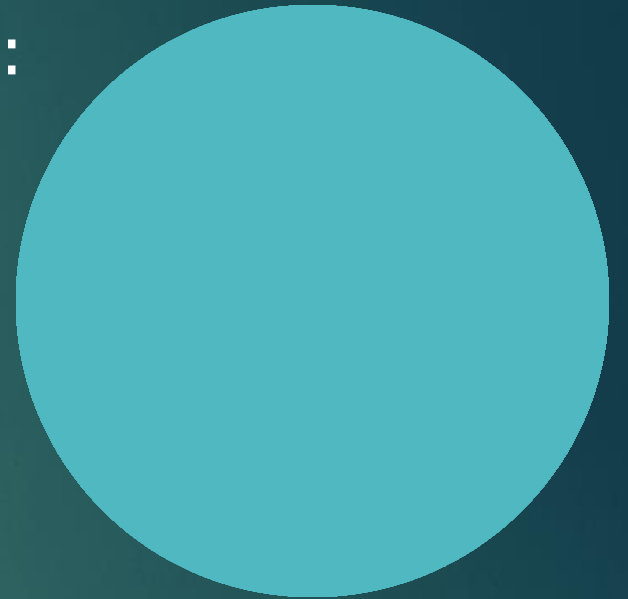
Stakeholders

Stakeholders are people or groups **affected by the project:**

- ▶ **Internal:** Project team, management
 - ▶ **External:** Clients, end users, regulatory bodies
- Stakeholder analysis helps understand and manage their **expectations, needs, and influence.**

Setting Objectives

- ▶ Projects should have **clear, measurable objectives**:
- ▶ **General vs. specific** goals
- ▶ Objectives should be **SMART**:
 - ▶ Specific
 - ▶ Measurable
 - ▶ Achievable
 - ▶ Relevant
 - ▶ Time-bound



The Business Case

The **business justification** for a project:

- ▶ Assesses whether the investment in a software project is worthwhile.
- ▶ Includes **cost-benefit analysis**, risk evaluation, alignment with strategic goals.

Project Success and Failure

- Success is more than just finishing on time and budget—it includes **meeting user needs**.

Common causes of failure:

- Poor requirement gathering
- Inadequate planning
- Lack of user involvement
- Uncontrolled changes
- Metrics for success include **time, cost, quality, user satisfaction, and business value**.

What is Management?

Defined as **coordinating people and resources** to achieve goals.

involves:

- planning – deciding what is to be done;
- organizing – making arrangements;
- staffing – selecting the right people for the job etc.;
- directing – giving instructions;
- monitoring – checking on progress;
- controlling – taking action to remedy hold-ups;
- innovating – coming up with new solutions;
- representing – liaising with clients, users, developer, suppliers and other stakeholders.

Management Control

- Focuses on **tracking progress**, identifying deviations, and taking **corrective actions**.

Key tools:

- **Milestones**
- **Gantt charts**
- **Performance metrics**
- Ensures project stays aligned with plan.

Traditional PM

Plan-driven (Waterfall)

Detailed upfront planning

Fixed scope, schedule, and budget

Control through documentation

Modern PM (Agile/Flexible)

Adaptive, iterative (Agile)

Continuous feedback and learning

Scope may evolve based on value

Collaboration and working software

Traditional vs. Modern Project Management Practices

MODERN PRACTICES
EMPHASIZE
COLLABORATION,
INCREMENTAL DELIVERY,
AND CUSTOMER
SATISFACTION.